

Assignment 1

Objective:

Perform a simple linear regression analysis to examine the relationship between **TV advertising budget** and **Sales** using the provided dataset.

Dataset:

Use the file: **tv_sales_data.csv**

Instructions:

1. **Import the dataset** into your programming environment.
2. **Fit a simple linear regression model:**

$$\text{Sales} = \beta_0 + \beta_1 \cdot \text{TV} + \epsilon$$

Report the following:

- Estimated values of **β_0** (intercept) and **β_1** (slope)
 - **Standard errors** of β_0 and β_1
 - **R-squared** value of the model
 - **p-values** for the intercept and slope
3. **Interpret the results:**
 - What does the sign and magnitude of β_1 indicate?
 - Based on the p-value, is there statistically significant evidence of a relationship between TV advertising and sales?
 - What proportion of the variance in sales is explained by the TV budget?

Submission:

Submit a short report (PDF or Word) or a notebook/script file with:

- Your code
- Output/results
- A brief interpretation (3–5 sentences)